

5 STICKS TOTAL



2669 Martin Luther King Jr Blvd
Youngstown, OH 44510

VALLOUREC STAR LP Seamless Tubular Products

Certified Test Report
05/17/2022
Revision: 0

Order: Y-221121981-001 Desc: Seamless Line Pipe- Normalized Rolled

Customer: Kelly Pipe Company

SLN / (Heat): NF0052

Grade: API 5L X42/B N/B/N PSL2 46th Edition May 01, 2019

Customer Order: 10P38957

Split: 00

OD(In): 4.500

Wall(In): 0.337

Lbs/Ft: 15.00

Length: Double Random

End Finish: Plain End Beveled

Comments: Melted and Manufactured in Youngstown, Ohio USA

Pipe is also manufactured to: ASTM A106B/C-19a, ASTM A53B-20, ASME SA106B/C-2021, ASME SA53B-2021. Vallourec Star's QA program meets the requirements of DIN EN10204-2004 Type 3.1. Satisfies NACE MR0175/ISO15156:2015 and ANSI/NACE MR0103/ISO17945:2015.

No repair welding has been conducted. Material is free from Mercury.

Mechanical Properties

			Tensile Size Specimen Cross Section			Strength (ksi)			Elongation		
Orientation	Type	Position	Width (In)	Thick (In)	Area (sqin)	Yield	Tensile	Y/T Ratio	Gage Length	% Elong	% RA
Longitudinal	Strip	Lead	1.000	0.340	0.3500	46.4	73.7	0.63	2	37.9	54.5

Required Min/Max Yield Strength (ksi): 42.1/71.8

Required Min/Max Tensile Strength (ksi): 70/95

Calculated Max Internal Yield (psi): 6081

Required Min Elongation(%): 30

Supplemental Requirements

Flattening Test	Passed
Hardness 3 Point	78.8 avg HRBW
Hydrostatic Pressure	4730 psi
Hydrostatic Hold Time	5 seconds

Chemical Analysis

Chem Spec: 61M

WT%	C	Mn	P	S	Si	Cu	Ni	Cr	Mo	Sn	Nb	V	Al	Ca	B	Ti	N	C.E.
Heat	0.21	0.70	0.009	0.006	0.25	0.23	0.07	0.10	0.02	0.009	0.000	0.001	0.032	0.0024	0.0001	0.003	0.0069	0.37
Product 1	0.20	0.71	0.007	0.005	0.25	0.24	0.07	0.09	0.02	0.008	0.000	0.000	0.031	0.0021	0.0001	0.003	0.0078	0.37
Product 2	0.21	0.71	0.007	0.005	0.25	0.24	0.07	0.09	0.02	0.008	0.000	0.000	0.030	0.0020	0.0001	0.003	0.0081	0.37

Charpy Impact Testing - Not required per API.

This material has been produced and tested in accordance with the requirements of applicable specifications unless otherwise listed below. We hereby certify that the above test results are representative of those contained in the records of the company. Any modification to this certification as provided by Vallourec Star without the expressed written consent of Vallourec Star negates the validity of this test report. Vallourec Star is not responsible for the inability of this material to meet specific applications.

THIS CERTIFICATE IS NOTARIZED ONLY WHEN REQUESTED.

Signed: *Jeff Schultz*

Name: Jeff Schultz, Lab Supervisor

Date Approved: 05/17/2022

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Inspection Company: Tuboscope

Non-Destructive Evaluation Criteria

EMI CHANNELS

ID

10% Longitudinal

10% Transverse

OD

10% Longitudinal

10% Transverse

WALL THICKNESS CHECK

75% Wall Reduction

87.5% Wall Reduction

Required Volumetric Coverage: 100%

END AREA

Mag Particle (Ref. Ind. Burmah/Castrol Strip)

OD

White Light

Longitudinal

ID

White Light

Longitudinal

Wall

Method: UT or Go/No-Go

25% Wall Thickness Verification

ADDITIONAL REQUIREMENTS

Notch Length 1.5" Max

Notch Width .020" Max

PSL2